K-Optimal and CS-Near-Optimal Exact Synthesis of Two-Qubit Clifford+CS Operators

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We present a method for decomposing a two-qubit Clifford+CS operator (equivalently, a 4×4 unitary with entries in $\mathbb{D}[i] = \mathbb{Z}[\frac{1}{2}, i]$) into a Clifford+CS circuit without ancillae. Given a unitary U, let k be the least number of K gates (i.e., scaled H gates) needed to implement it. Our method decomposes U, up to a global phase, into a circuit of length at most $10k+9$, using exactly k K gates (i.e., K-optimal) and at most $k+1$ CS gates. We prove that the number of CS gates used is at most 2 times optimal (i.e., CS-near-optimal). We also improve the lower bound of the optimal CS-count in [17] by a factor of 2. Let l be the least natural number such that the entries of $(1+i)^l U$ are all Gaussian integers; we prove that $k = 2l - \{0, 1, 2\}$. In experiments, our method always achieves the optimal CS-count and yields a mean K-count reduction of 19% compared to the best known result. Part of our approach is similar to that of Russell [34] for the Clifford+T gate set, but provides explicit bounds (e.g., $10k+9$ vs. $O(k)$; the two k's are different but related) and operates over the simpler ring $\mathbb{Z}[\frac{1}{2}, i]$, a subring of the ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$ used by Russell.

I. INTRODUCTION

In quantum computation, decomposing unitary operators into gates from a fixed universal set is a fundamental problem [32]. The Clifford+Controlled-S gate set (abbreviated Clifford+CS) is universal, just like the well-known Clifford+T gate set. We study the exact synthesis of two-qubit Clifford+CS operators. Previous work on two-qubit synthesis of Clifford+CS or Clifford+T operators has mainly focused on minimizing the CS- or T-count, since these gates are more expensive than Clifford operators in certain scenarios. However, work on T-gate optimization [1, 5, 9, 10, 20, 24] has suggested that the H gate is a bottleneck for T-optimization methods. Quoting from [1]: "It has been shown that T -depth optimizations can be implemented efficiently for circuits consisting only of T and CNOT gates and that H gates impede the optimization significantly"

We propose a method that produces circuits with a minimal number of K (i.e., scaled H) gates and a near-minimal number of CS gates, thereby mitigating this bottleneck. The problem of simultaneously minimizing two different gate counts is also of mathematical interest: previous work on single-qubit Clifford+T synthesis achieved the simultaneous minimum use of the Hadamard and T gates [14, 26, 30], and extending this to the multi-qubit case has been an open problem.

A. Background

Consider the following unitary operators:

$$i, K = \omega^7 H = \frac{1}{1+i}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}, S = \begin{bmatrix}1 & 0\\ 0 & i\end{bmatrix}, \text{ and}$$

$$CZ = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -1\end{bmatrix}.$$

Here, i is the complex unit; $\omega = e^{\frac{i\pi}{4}}$ is a primitive 8th root of unity; H is the Hadamard gate and K is a scaled version of H; S is sometimes called the *phase gate*; and CZ is the controlled-Z gate. Under multiplication, identities, and tensor products, these operators generate the *Clifford group*. (The precise group depends on which scalars are included; we include only the scalars ± 1 and $\pm i$, so the above gates generate the Clifford group exactly.)

Every operator U obtained in this way has size $2^n \times 2^n$ for some natural number n, and we say that U acts on n qubits. The group of n-qubit Clifford operators is finite for any given n (see, e.g., [35]), and therefore not universal for quantum computing. Universality is obtained by adding the controlled phase gate

$$CS = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & i\end{bmatrix}.$$

The resulting operators are called Clifford+CS operators. Note that $CZ = CS^2$. We write $\mathcal{CCS}$ for the group of two-qubit Clifford+CS operators and $\mathcal{CIR}$ for the set of two-qubit Clifford+CS circuits (a circuit being a finite sequence of gates). Each circuit $D \in \mathcal{CIR}$ implements a unique operator in $\mathcal{CCS}$, denoted $[D]$. That is, $[\_]$ is the matrix semantics function from $\mathcal{CIR}$ to $\mathcal{CCS}$.

We use the following notation: $K_0 = K \otimes I$, $K_1 = I \otimes K$, and similarly for S_0 and S_1, where I is the 2×2 identity operator. We identify the scalar i with the 4×4 matrix $i\,(I \otimes I)$. For controlled gates, the last index denotes the target. For example, CX_{01} is the controlled-X gate with control qubit 0 and target qubit 1, shortened to CX; CX_{10} is shortened to XC. The controlled-

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S and controlled-Z gates are symmetric: $CS_{jk} = CS_{kj}$ and $CZ_{jk} = CZ_{kj}$, so we simply write CS and CZ. We also use the Z-gate, the X-gate, the controlled-K gate $CK = I \oplus K$, and the swap gate Ex. These are related to the primitive gates by $Z = S^2$, $X = KZK \cdot i$, $CX = K_1 CZ K_1 \cdot i, Ex = CX\, XC\, CX$, and

$$CK = CZ\, Z_0 S_0 Z_1 S_1 K_1 CS K_1 S_1 \cdot i. \tag{1}$$

By Equation (1), $CK \in \mathcal{CCS}$. As with CX, we shorten CK_{01} to CK and CK_{10} to KC.

B. Related work

Operator synthesis can be performed exactly or approximately. We restrict our attention to exact synthesis and list the most closely related works.

1. Single-qubit and single-qudit

For single-qubit Clifford+T synthesis, the works [14, 26, 30] achieved the simultaneous minimum use of both the Hadamard and T gates. The works [16, 33] achieved the minimum T-count for single-qutrit Clifford+T synthesis. The work [22] considered single-qudit Clifford+T synthesis. Forest et al. [12] generalized exact synthesis from Clifford+T to single-qubit Clifford-cyclotomic gate sets. Kliuchnikov and Yard [28] developed a general algebraic framework for exact synthesis over number rings, unifying several gate sets under one theory.

2. Two-qubit

Three works on two-qubit synthesis guarantee optimality of a non-Clifford gate: Glaudell et al.'s CS-optimal synthesis of Clifford+CS operators [17] and Li et al.'s T-near-optimal synthesis of Clifford+T operators [29], whose worst-case T-count is at most 10 times optimal. Neither considered H-count optimization, but for the former, we can deduce that its H-count is at most 2 times optimal. Glaudell et al. [15] later gave a high-performance exact synthesis framework for two-qubit Clifford+T circuits that optimizes the T-count exactly, using a meet-in-the-middle strategy with algebraic canonicalization and lookup tables. All three of these works rely on the exceptional isomorphism $SU(4) \cong \mathrm{Spin}(6)$. By contrast, our work does not use this isomorphism. Instead, we employ a residue analysis similar to that of Russell [34], but over the simpler ring $\mathbb{Z}[\frac{1}{2}, i] \subset \mathbb{Z}[\frac{1}{\sqrt{2}}, i]$. Given a unitary U, Russell's method outputs a circuit of gate count $O(k)$, where k is the least natural number such that the entries of $(1+\omega)^k U$ lie in $\mathbb{Z}[\omega]$.

3. Multi-qubit and multi-qutrit

Multi-qubit Clifford+T synthesis was introduced by Giles and Selinger [13], whose algorithm has worst-case gate count $O\left(3^{2^n} nk\right)$. This was later improved [25] to $O\left(4^n nk\right)$. Glaudell et al. gave exact multi-qubit synthesis for several subclasses of Clifford+T operators [4], such as the Toffoli+Hadamard operators. Amy et al. adapted the method of [25] to Toffoli+Hadamard operators, achieving a similar improvement in gate count [3]. None of these multi-qubit synthesis works consider T-count optimality. Kliuchnikov and Schönnenbeck [27] recently developed an optimal search algorithm for multi-qubit exact synthesis using A^* search; for two-qubit circuits, its runtime is proportional to the T-count. The works [2, 11] considered exact synthesis of multi-qubit Clifford+Cyclotomic operators, while [19, 23] treated the multi-qutrit case. The work [18] proved that qutrit metaplectic gates are a subset of Clifford+T. The work [31] proposed a heuristic algorithm achieving V-gate optimality.

II. SOME ALGEBRA

A. Rings

We write $\mathbb{N} = \{0, 1, \ldots\}$ for the natural numbers and $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$ for the integers, real numbers, and complex numbers respectively. We write $z^\dagger$ for the complex conjugate of a complex number z.

Let $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ be the ring of Gaussian integers. It is a Euclidean domain [6], so division with remainder, greatest common divisors, divisibility, and the notion of a prime (i.e., a non-unit that cannot be written as a product of two non-units) are all available in $\mathbb{Z}[i]$. Let $\gamma = 1+i$. It is well known that γ is a Gaussian prime [21]. Since $\gamma^2 = 2i$, the elements γ^2 and 2 divide each other. Using Euclidean division, one can verify that

$$\mathbb{Z}[i]/(\gamma) = \{0, 1\}, \quad \mathbb{Z}[i]/(2) = \mathbb{Z}[i]/(\gamma^2) = \{0, 1, i, 1+i\},$$

$$\text{and} \quad \mathbb{Z}[i]/(\gamma^3) = \{\pm 1, \pm i, 0, 1+i, 1-i, 2\}.$$

The above is a slight abuse of notation: strictly, $\mathbb{Z}[i]/(\gamma)$ denotes the quotient ring of $\mathbb{Z}[i]$ modulo the ideal (γ), whose elements are residue classes $x \bmod (\gamma)$ for $x \in \mathbb{Z}[i]$. We have omitted $\bmod (\gamma)$ for brevity. The sets $\{0, 1\}$ etc. are complete sets of residue class representatives.

Let $\mathbb{D} = \mathbb{Z}[\frac{1}{2}] = \left\{\frac{a}{2^k} \mid a \in \mathbb{Z}, k \in \mathbb{N}\right\}$ be the ring of *dyadic fractions*, i.e., fractions whose denominator is a power of 2. Consider the ring $\mathbb{D}[i] = \{r + si \mid r, s \in \mathbb{D}\} = \mathbb{Z}[\frac{1}{2}, i]$. Every $t \in \mathbb{D}[i]$ satisfies $\gamma^k t \in \mathbb{Z}[i]$ for some $k \in \mathbb{N}$, motivating the following definition.

Definition II.1. Let $t \in \mathbb{D}[i]$. A natural number $k \in \mathbb{N}$ is called a *denominator exponent* for t if $\gamma^k t \in \mathbb{Z}[i]$. The least such k is called the *least denominator exponent* of t, denoted by $\mathrm{lde}(t)$. Equivalently, the least denominator exponent of t is the smallest $k \in \mathbb{N}$ such that t can be written in the form $\frac{s}{\gamma^k}$, where $s \in \mathbb{Z}[i]$.

More generally, we say that k is a denominator exponent for a vector or matrix if it is a denominator exponent for all of its entries. The least denominator exponent for a vector or matrix is therefore the least k that is a denominator exponent for all of its entries.

Remark II.2. By enumeration, the lde of any 2-qubit Clifford operator is at most 2.

B. Residue rings

The following characterization of the quotient rings is well known:

$$\mathbb{Z}[i]/(\gamma^n) = \left\{ \sum_{i=0}^{n-1} b_i \gamma^i \bmod (\gamma^n) \,|\, b_i \in \{0,1\} \right\}.$$

We give an elementary proof by induction on n; a similar argument for the case $\mathbb{Z}[\sqrt{2}]/(\sqrt{2}^n)$ appears in Clement [8]. The base case $n = 1$ is immediate. Suppose the claim holds for n, i.e., every $x \in \mathbb{Z}[i]$ is congruent to some $\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n)$, meaning

$$x = \sum_{i=0}^{n-1} b_i \gamma^i + k\gamma^n \text{ for some } k.$$

Since $k = b_n + k'\gamma$ for some $b_n \in \{0, 1\}$, substituting and simplifying gives

$$x = \sum_{i=0}^{n} b_i \gamma^i + k'\gamma^{n+1}, \text{ i.e., } x \equiv \sum_{i=0}^{n} b_i\gamma^i \bmod (\gamma^{n+1}).$$

For brevity, we will omit the $\bmod (\gamma^n)$, and since $b_i \in \{0, 1\}$, we write the binary string $(b_0b_1 \dots)_2$ for the sum; the parentheses and subscript $_2$ may be omitted when the context makes clear that $b_0b_1 \dots$ denotes a binary residue.

Moreover, residue classes are distinct: if $b_0b_1 \dots b_{n-1}$ and $b'_0b'_1 \dots b'_{n-1}$ are congruent, i.e., $\sum_{i=1}^{n-1} b_i\gamma^i - \sum_{i=1}^{n-1} b'_i\gamma^i = k\gamma^n$ for some k, then $b_i = b'_i$ for all i. Indeed, reducing both sides modulo (γ) gives $b_0 = b'_0$. Cancelling b_0 and b'_0 and dividing by γ yields

$$\sum_{i=1}^{n-1} b_i\gamma^{i-1} - \sum_{i=1}^{n-1} b'_i\gamma^{i-1} = k\gamma^{n-1}.$$

Reducing modulo (γ) again gives $b_1 = b'_1$. Repeating this argument shows $b_i = b'_i$ for all i.

We write $\rho_n(x)$ for the canonical projection from $\mathbb{Z}[i]$ to n-digit residues. The proof above shows that $\rho_n(x)$ is *stable* under increase of n: the first n digits of $\rho_{n+1}(x)$ coincide with $\rho_n(x)$. Moreover, ρ_2 is stable under complex conjugation: $\rho_2(x^\dagger) = \rho_2(x)$. We sometimes write $x \equiv_{\gamma^n} y$ as shorthand for $x \equiv y \bmod \gamma^n$. Modulo γ^3, we have

x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$
1	100	-1	101	i	111	$-i$	110
0	000	$1+i$	010	$1-i$	011	2	001

The (mod γ^n) residue classes form a commutative ring. Modulo γ^2, the ring addition coincides with bitwise exclusive or (denoted $\oplus$); $(10)_2$ is the multiplicative identity; $(00)_2$ is the zero element; and multiplication by $(01)_2$ is the right shift operation. By distributivity, $11 \cdot b_0b_1 = 10 \cdot b_0b_1 + 01 \cdot b_0b_1 = b_0(b_1 \oplus b_0)$. The additive inverse of an element is itself. Since $\rho_2(i) = 11$, whenever $\rho_1(x) = 1$, either $\rho_2(x) = 10$ or $\rho_2(ix) = 10$. Similarly, modulo γ^3, the addition rule is

$$b_0b_1b_2 + b'_0b'_1b'_2 = (b_0 \oplus b'_0)(b_1 \oplus b'_1)(b_2 \oplus b'_2 \oplus (b_0 \cdot b'_0)),$$

where $\cdot$ denotes multiplication in $\mathbb{Z}[i]/(\gamma)$, equivalently the logical "and" of two bits. The multiplicative identity is 100 and the zero element is 000. Multiplication by 010 is a right shift by 1 digit and multiplication by 001 is a right shift by 2 digits. The rest of the multiplication table follows by distributivity. The additive inverse is

$$-(b_0b_1b_2) = b_0b_1(b_2 \oplus b_0).$$

Inspecting the ρ_3 table, we see that ρ_3 is not stable under complex conjugation; instead, if $\rho_3(x) = abc$, then $\rho_3(x^\dagger) = ab(c \oplus b)$. Since $\rho_3(i) = 111$, whenever $\rho_1(x) = 1$, there exists $k \in \{0, 1, 2, 3\}$ such that $\rho_3(i^kx) = 100$.

Definition II.3. Let l be a denominator exponent (not necessarily the least) for $x \in \mathbb{D}[i]$. We define the residue function $\rho^l_n(x) = \rho_n(\gamma^l x)$. We call $\rho^l_n(x)$ the (n, l)-residue of x. If l is a denominator exponent for a vector or matrix A, $\rho^l_n(A)$ is defined entry-wise.

Definition II.4. For $x \in \mathbb{Z}[i]$, if $\rho_1(x) = 0$, we say x is *even*, otherwise *odd*.

For example, $1 + 2i$ is odd, while $1 + 3i$ is even. In our convention, an even Gaussian integer is one divisible by γ, not necessarily by 2. However, for $x \in \mathbb{Z}$, x is even (resp. odd) as an integer if and only if it is even (resp. odd) as a Gaussian integer, so our definition extends the usual notion of parity on the integers. The following lemmas are straightforward.

Lemma II.5. *Let $\alpha = a + bi \in \mathbb{Z}[i]$. Then α is odd if and only if $\|\alpha\|^2 = a^2 + b^2$ is an odd integer.*

Lemma II.6. *If $\alpha, \beta \in \mathbb{Z}[i]$ satisfy $\alpha\gamma \equiv_{\gamma^{n+1}} \beta\gamma$, then $\alpha \equiv_{\gamma^n} \beta$.*

Lemma II.7. *For $t \in \mathbb{D}[i]$, if $k = \mathrm{lde}(t) > 0$, then $\gamma^k t$ is odd.*

Lemma II.8. *The lde function on $\mathbb{D}[i]$ and matrices with $\mathbb{D}[i]$ entries is subadditive, i.e., for $x, y \in \mathbb{D}[i]$ and matrices A, B with $\mathbb{D}[i]$ entries*

$$\operatorname{lde}(xy) \leq \operatorname{lde}(x) + \operatorname{lde}(y),\ \operatorname{lde}(AB) \leq \operatorname{lde}(A) + \operatorname{lde}(B).$$

Moreover, if x, y are both odd, then lde is additive, i.e., the inequality becomes equality.

C. Action on residues

Suppose $\rho_2(x) = b_0 b_1$; then $\rho_2(ix) = b_0(b_1 \oplus b_0)$. We say i acts on $b_0 b_1$ by sending it to $b_0(b_1 \oplus b_0)$. Given an n-digit residue $b_0 b_1 \ldots b_{n-1}$, a *right shift* produces the $(n{+}1)$-digit residue $0b_0 b_1 \ldots b_{n-1}$. Given an $(n{+}1)$-digit residue $0b_0 b_1 \ldots b_{n-1}$, a *left shift* produces $b_0 b_1 \ldots b_n$. We only allow shifts that are "well-defined" in the sense that no information is lost and no unknown digits are introduced. (The usual right shift was used above when describing multiplication in $\mathbb{Z}[i]/(\gamma^2)$ and $\mathbb{Z}[i]/(\gamma^3)$.) We write $LS(x)$ for the left shift of x and RS for the right shift.

Lemma II.9. *For $t \in \mathbb{D}[i]$, k a denominator exponent of t, we have*

$$\rho^{k+1}_{n+1}(t) = \rho^k_{n+1}(\gamma t) = RS(\rho^k_n(t)),$$

and if $\gamma^k t$ is even,

$$\rho^{k-1}_n(t) = \rho^k_n(t/\gamma) = LS(\rho^k_{n+1}(t)).$$

Given $x, y \in \mathbb{D}[i]$ with lde $k > 0$, we have $\rho^k_2(x) = 1b$ and $\rho^k_2(y) = 1b'$ for some $b, b' \in \{0, 1\}$. Hence $\rho^k_2(x{+}y) = 1b{+}1b' = 0(b{\oplus}b')$, and so $\rho^{k-1}_1(x + y) = LS(\rho^k_2(x{+}y)) = b \oplus b'$. Similarly, $\rho^{k-1}_1(x - y) = LS(\rho^k_2(x - y)) = LS(1b - 1b') = LS((1-1)(b - b')) = b \oplus b'$, since subtraction in $\mathbb{Z}/(2)$ coincides with $\oplus$ (every element is its own additive inverse). If $b \oplus b' = 1$, then $\operatorname{lde}(x \pm y)$ decreases by 1; if $b \oplus b' = 0$ and $k \geq 2$, then $\operatorname{lde}(x \pm y)$ decreases by at least 2. Since dividing t by γ increases the lde by 1, for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho^k_3(x) = 10b$ and $\rho^k_3(y) = 10b'$, we obtain

$$\rho^k_2(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho^k_2(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho^k_3(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} 0(b \oplus b' \oplus 1) \\ 0(b \oplus b') \end{bmatrix}.$$

Hence $\operatorname{lde}(K \begin{bmatrix} x \\ y \end{bmatrix}) = k - 1$, with one entry having lde $k-1$ and the other at most $k-2$ (provided $k \geq 2$).

Remark II.10. Note that K acting on a pair of entries with the same lde $k > 0$ and proper residues can only decrease the lde of the pair by at most 1. Also, the K gate is the only gate in the Clifford+CS operator generating set that affects the lde. These two facts are needed when proving the optimality of our method.

P	circuit	P	circuit	P	circuit
0123	identity	1023	$X_1 \cdot CX$	2103	$X_0 \cdot XC$
1203	$Ex \cdot X_0 \cdot XC$	2013	$Ex \cdot X_1 \cdot CX$	0213	Ex
3210	$X_1 \cdot X_0$	2310	$X_0 \cdot CX$	2130	$Ex \cdot X_1 \cdot XC$
3120	$Ex \cdot X_1 \cdot X_0$	1320	$Ex \cdot X_0 \cdot CX$	1230	$X_1 \cdot XC$
3012	$XC \cdot X_1$	0312	$Ex \cdot XC$	0132	CX
3102	$Ex \cdot CX \cdot X_0$	1302	$Ex \cdot X_0$	1032	X_1
3021	$Ex \cdot XC \cdot X_1$	0321	XC	0231	$Ex \cdot CX$
3201	$CX \cdot X_0$	2301	X_0	2031	$Ex \cdot X_1$

TABLE I. Circuit implementation of 24 permutation matrices.

In general for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho^k_3(x) = 1b_0 b_1$ and $\rho^k_3(y) = 1b'_0 b'_1$,

$$\rho^k_2(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} (b_0 \oplus b'_0)(b_1 \oplus b'_1 \oplus 1) \\ (b_0 \oplus b'_0)(b_1 \oplus b'_1) \end{bmatrix}.$$

If $\rho^k_3(x) = 0b_0 b_1$ and $\rho^k_3(y) = 0b'_0 b'_1$,

$$\rho^k_2(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} (b_0 \oplus b'_0)(b_1 \oplus b'_1) \\ (b_0 \oplus b'_0)(b_1 \oplus b'_1) \end{bmatrix}.$$

If $\rho^k_3(x) = 1b_0 b_1$ and $\rho^k_3(y) = 0b'_0 b'_1$,

$$\rho^{k+1}_3(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b'_0)(b_1 \oplus b'_1) \\ 1(b_0 \oplus b'_0)(b_1 \oplus b'_1) \end{bmatrix}.$$

If $\rho^k_3(x) = 0b_0 b_1$ and $\rho^k_3(y) = 1b'_0 b'_1$,

$$\rho^{k+1}_3(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b'_0)(b_1 \oplus b'_1) \\ 1(b_0 \oplus b'_0)(b_1 \oplus b'_1 \oplus 1) \end{bmatrix}.$$

III. SPECIAL UNITARIES

A. Permutation matrices

Let $a_0 a_1 a_2 a_3$ be a permutation of $\{0, 1, 2, 3\}$. The permutation matrix $P_{a_0 a_1 a_2 a_3}$ is the unitary matrix that maps e_i to e_{a_i}, where $\{e_i\}$ is the standard basis. All 24 permutation matrices are Clifford operators and can be implemented by circuits of length at most 3; see Table I.

B. Diagonal unitary matrices

The only diagonal unitary matrices with entries in $\mathbb{D}[i]$ and $\mathbb{Z}[i]$ are of the form

$$\operatorname{diag}(i^a, i^b, i^c, i^d),\ \text{where } a, b, c, d \in \{0, 1, 2, 3\}.$$

They can all be implemented by Clifford+CS circuits of length at most 6, up to a global phase i^k, using at most 1 CS gate. This is because

$$Z_0^{b_0} Z_1^{b_1} S_0^{b_2} S_1^{b_3} CZ^{b_4} CS^{b_5} \cdot i^k,\ b_i \in \{0,1\}\,,\ k \in \{0,1,2,3\}\,,$$

are all distinct and diagonal, and their number is 4^4, which equals the cardinality of the set of diagonal unitary matrices.

C. Generalized permutation unitary matrices

A *generalized permutation* (unitary matrix with entries in $\mathbb{D}[i]$) is a matrix having the same nonzero pattern as a permutation matrix, but whose nonzero entries may be arbitrary powers of i (i.e., unit-length elements of $\mathbb{Z}[i]$). Every 4×4 generalized permutation is a product of a permutation matrix and a diagonal unitary matrix, and hence can be implemented by a circuit of length at most 9 containing at most 1 CS gate and no K gate. The group of generalized permutations is a semidirect product of the permutation group and the diagonal unitary group. In particular, for any permutation matrix P and diagonal matrix D, there exists a diagonal matrix D' such that

$$PD = D'P$$

and the CS-count (see Definition V.1) of D and D' are the same.

There are finitely many generalized permutation Clifford+CS operators, and each has lde 0. Conversely, every lde 0 Clifford+CS operator is a generalized permutation. This is because a unit vector with lde 0 can only have the form $(i^k,0,0,0)^T$ for $k \in \{0,1,2,3\}$, up to a permutation of rows.

IV. THE EXACT SYNTHESIS ALGORITHM

The method in this section follows a residue analysis similar to Russell's [34] for the ring $\mathbb{Z}[\frac{1}{\sqrt{2}},i]$, but operates over the simpler subring $\mathbb{Z}[\frac{1}{2},i]$. Similar residue analysis methods are widely used in exact synthesis; see, e.g., [3, 13, 14, 17, 29]. Compared to the ad hoc use of residues in prior work, we have given in Section II a uniform treatment based on binary residues, which we conjecture generalizes to any ξ-ring as defined in [27].

A. Residue pattern

We begin with several lemmas. Since $\mathbb{Z}[i]$ and $\mathbb{D}[i]$ are subrings of the complex numbers, the standard notions of complex conjugation (denoted $\_^\dagger$), inner product, norm, transposition (denoted $\_^T$), and conjugate transposition (also denoted $\_^\dagger$) carry over. For a vector $v \in \mathbb{D}[i]^n$, the *norm* is $\|v\| = \sqrt{v^\dagger v}$, and $\|v\|^2 = v^\dagger v \in \mathbb{D}$ since $v^\dagger v \in \mathbb{D}[i] \cap \mathbb{R}$. Similarly, if $v \in \mathbb{Z}[i]^n$, then $v^\dagger v \in \mathbb{Z}$. A *unit vector* is a vector of norm 1. For a vector v, the notation v_j refers to its jth entry; entries are numbered starting from $j=0$.

Lemma IV.1. *For any $A \in \mathcal{CCS}$ with lde l, there exist permutation matrices L and R such that $\rho_1^l(LAR)$ or $\rho_1^l(LAR)^T$ equals one of the following*

$$(i)\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix},\quad (ii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (iii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{bmatrix},$$

$$(iv)\begin{bmatrix}1&1&1&1\\1&1&1&1\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (v)\begin{bmatrix}1&1&0&0\\1&1&0&0\\1&1&1&1\\1&1&1&1\end{bmatrix},\quad (vi)\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}.$$

We say A has such a pattern *and write* pat(A) *for the pattern of A. Moreover, A has pattern (i) if and only if* lde$(A) = 0$*, i.e., when A is a generalized permutation. Patterns (iv), (v), and (vi) occur only when* lde$(A) > 1$*. Each Clifford operator has pattern (i), (iii), or (vi).*

Note that all cases except for Case (iv) are transposition-invariant, e.g., (vi)^T = (vi), up to permutation of rows and columns. Before proving this lemma, we prove several useful lemmas.

Lemma IV.2. *Let v be a unit vector in $\mathbb{D}[i]^4$ with lde $k > 0$. There exists a generalized permutation matrix P such that $\rho_1^k(Pv) = (1,1,0,0)^T$ or $(1,1,1,1)^T$, $\rho_2^k(Pv) = (10,10,01,00)^T$ or $(10,10,10,10)^T$ when $k > 1$, and $\rho_2^1(Pv) = (10,10,00,00)^T$ when $k = 1$.*

Proof. Let $w = \gamma^k v \in \mathbb{Z}[i]^4$. We first show that w has an even number of odd entries. We have

$$\sum_{j=0}^{3} \|w_j\|^2 = \|w\|^2 = |\gamma|^{2k}\|v\|^2 = (\gamma^\dagger\gamma)^k \equiv 0 (\mathrm{mod}\ \gamma).$$

Therefore, an even number of indices $j \in \{0,\dots,3\}$ satisfy $\|w_j\|^2$ odd. By Lemma II.5, an even number of the w_j are odd.

We can then find a permutation matrix P such that $\rho_2^k(Pv) = (1a,1b,0c,0d)^T$ or $(1a,1b,1c,1d)^T$. In the latter case, multiplying by a suitable diagonal matrix (i.e., multiplying individual entries by appropriate powers of i) makes all second digits zero, yielding $(10,10,10,10)^T$.

In the former case, multiplying by a suitable diagonal matrix makes the second digits of the odd entries zero, yielding $(10,10,0c,0d)^T$. Applying X_0CKX_0 (equivalently, applying K to the pair of odd entries) produces $(00,01,0c,0d)^T$ or $(01,00,0c,0d)^T$. Since multiplication by unitaries preserves unit vectors, this is the residue of a unit vector with lde $k-1$. When $k > 1$, its $(1,k-1)$-residue must have the form $(1,1,0,0)^T$ up to a permutation, so $c = 1$ and $d = 0$.

When $k=1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ for $l, m \in \{0,1,2,3\}$, up to a permutation. $\square$

Lemma IV.3. *Let u, v be unit vectors in $\mathbb{D}[i]^4$ with lde $k>0$ such that $u \perp v$, i.e., $u^\dagger v = 0$. Then there exists a generalized permutation matrix Q such that $\rho_1^k(Q[u|v])$ equals one of*

$$\begin{bmatrix}1&1\\1&1\\0&0\\0&0\end{bmatrix}, \begin{bmatrix}1&0\\1&0\\0&1\\0&1\end{bmatrix}, \begin{bmatrix}1&1\\1&1\\1&0\\1&0\end{bmatrix}, \begin{bmatrix}1&1\\1&1\\0&1\\0&1\end{bmatrix}, \text{ or } \begin{bmatrix}1&1\\1&1\\1&1\\1&1\end{bmatrix}.$$

Proof. By Lemma IV.2, each column can have only an even number of 1s. Up to a permutation of rows, the only missing case is

$$\begin{bmatrix}1&0\\1&1\\0&1\\0&0\end{bmatrix}.$$

Then $\rho_1^k(u^\dagger v) = \sum_{j=0}^{3} \rho_1^k(u_j^\dagger v_j) = 1$, contradicting $\rho_1^k(u^\dagger v) = \rho_1^k(0) = 0$. In other words, the number of indices at which both u and v have $(1,k)$-residue equal to 1 must be even. $\square$

Proof of Lemma IV.1. The rows and columns of a unitary matrix are unit vectors, and all row (column) vectors are pairwise orthogonal. Lemmas IV.2 and IV.3 force the pattern to be (ii)–(vi) when $k>0$. When $k=0$, the only unitaries with lde 0 are the generalized permutations, which have pattern (i). When $k=1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ up to a permutation, so the residue cannot have four 1s in a column. Hence patterns (iv), (v), and (vi) occur only when $\mathrm{lde}(A) > 1$. By enumeration, each Clifford operator has pattern (i), (iii), or (vi). $\square$

B. Transitions between patterns

Lemma IV.4. *For any $A \in \mathcal{CCS}$ with lde $k>0$, there exist generalized permutations L, R, L', R' such that one of the following holds:*

- $\mathrm{lde}(K_1 LAR) \le k-1$;
- $\mathrm{lde}(LARK_1) \le k-1$;
- $\mathrm{lde}(L'K_1LARR'K_1) \le k-1$;
- $\mathrm{lde}(K_1L'K_1LARR') \le k-1$;
- $\mathrm{lde}(L'CKLAR) \le k-1$ *(only when* $\mathrm{lde}(A)=1$*).*

Each inequality is in fact an equality, since by Remark II.10, K acting on a pair of entries decreases the lde by at most 1 (for the cases with two K gates, only the outer K decreases the lde; see the proof). The proof proceeds by case analysis on $\rho_1^k(A)$. Since $k>0$, Case (i) is excluded, so we begin with Case (ii).

1. Case ii

If $k>1$, by Lemma IV.2, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix}10&10&01&00\\10&1a&0b&0\bar{b}\\01&0c&0d&0e\\00&0\bar{c}&0f&0g\end{bmatrix},$$

where $\bar{b} = b \oplus 1$ and b,c,d,e,f,g are indeterminates (henceforth we use indeterminates in residue matrices without explicit declaration). We claim $a=0$ and $b=c=1$. Taking the inner product of the first two columns gives

$$10 + 1a + 00 + 00 = 00 \implies a = 0.$$

The inner product of the second and third rows gives $c=1$; similarly, $b=1$. Now $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix}10&10&01&00\\10&10&01&00\\01&01&0d&0e\\00&00&0f&0g\end{bmatrix}.$$

Moreover, since $k>1$ and the third and fourth rows (columns) have lde $k-1>0$, Lemma IV.2 gives $d=f$, $d=e$, and $e=g$, so $d=e=f=g$. Now $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix}10&10&01&00\\10&10&01&00\\01&01&0d&0d\\00&00&0d&0d\end{bmatrix}.$$

Applying K_1S_1 on the left reduces this to Case (iv).

If $k=1$, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix}10&10&00&00\\10&1a&00&00\\00&00&00&00\\00&00&00&00\end{bmatrix}.$$

As before, $a=0$. Applying $X_0 \cdot CK \cdot X_0$ on the left decreases the lde to 0, reaching pattern (i).

Remark IV.5. The two generalized permutations used to refine the pattern can be chosen to be CS-free. This is because the $(2,k)$-residues of the two rows (columns) with even entries are stable under multiplication by i, i.e., we have the liberty of updating the two generalized permutations by a factor of CS gate without affecting their purpose. Also, if a generalized permutation P is implementable using one CS gate, then $P \cdot CS$ is implementable using no CS gate.

2. Case iii

If $k > 1$, by Lemma IV.2, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 0b & 0\bar{b} \\ 01 & 0c & 10 & 10 \\ 00 & 0\bar{c} & 10 & 1d \end{bmatrix}.$$

The inner product of the first two columns gives $a = 0$. The inner product of the second and third rows gives $c = 0$; similarly, $b = 0$. Now $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 01 & 00 & 10 & 10 \\ 00 & 01 & 10 & 10 \end{bmatrix}.$$

Applying $K_1 S_1$ on the left reduces this to Case (vi).

If $k = 1$, by Lemma IV.2, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 10 & 10 \\ 00 & 00 & 10 & 1d \end{bmatrix}.$$

As before, $a = 0$ and $d = 0$. Applying K_1 on the left decreases the lde to 0, reaching pattern (i).

3. Case iv

This case occurs only when $k > 1$. By Lemma IV.2, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1b & 1c \\ 01 & 0d & 0e & 0f \\ 00 & 0\bar{d} & 0\bar{e} & 0\bar{f} \end{bmatrix}.$$

The inner product of the first two columns gives $a = 0$; similarly, $b = c = 0$. Since the third row has lde $k-1 > 0$, either $d = e = f = 1$ or exactly one of d, e, f is 1 and the rest are 0. In the latter case, up to a column permutation, we may assume $d = 1$ and $e = f = 0$. In both cases, right multiplication by K_1 decreases the lde, and moreover, up to permutations of rows and columns, reaching $(1, k-1)$-residue

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ p & \bar{p} & q & \bar{q} \\ x & x & y & y \\ z & z & w & w \end{bmatrix}.$$

Since $k - 1 > 0$, taking the inner product of the first two columns gives $x = z$; similarly, $y = w$. Then $p = 1$ since each column (row) has an even number of odd entries; similarly, $q = 1$. Taking the inner product of the first and third rows gives $x = y$. Taking the inner product of the first two rows gives $p = q$. Depending on whether $x = 0$ or 1, the target pattern is (ii) or (v).

Remark IV.6. As in Remark IV.5, the left generalized permutation can be chosen to be CS-free.

4. Case v

This case occurs only when $k > 1$. By Lemma IV.2, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 00 & 01 \\ 10 & 1b & 10 & 1f \\ 10 & 1c & 1e & 10 \end{bmatrix}.$$

We claim $a = 0$ and $b = c = e = 1$. The inner product of the first two rows gives $a = 0$. The inner product of the first and third rows gives $b = 1$; similarly, $c = 1$. The inner product of the second and third columns gives $e = 1$; similarly, $f = 1$. Now the pattern is

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 10 & 11 & 10 & 11 \\ 10 & 11 & 11 & 10 \end{bmatrix}.$$

Applying K_1 on the left reduces this to Case (iv).

5. Case vi

This case occurs only when $k > 1$. By Lemma IV.2, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1d & 1g \\ 10 & 1b & 1e & 1h \\ 10 & 1c & 1f & 1j \end{bmatrix}.$$

The inner product of the first two columns shows that a, b, c cannot all be 1. Up to a row permutation, we may assume $a = 0$, which gives $b = c$; similarly, $d = g$.

If $b = c = 1$ and $d = g = 1$, then $\mathrm{pat}(A)$ is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1h \\ 10 & 11 & 1f & 1j \end{bmatrix}.$$

The inner product of the first and third columns gives $f=\bar{e}$; similarly, $h=\bar{e}$ and $j=\bar{f}$. Now $\text{pat}(A)$ is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1\bar{e} \\ 10 & 11 & 1\bar{e} & 1e \end{bmatrix}.$$

We show this pattern is impossible by examining the $(3,k)$-residue. Up to multiplication by powers of -1 on appropriate rows and columns (achieved by multiplication by two appropriate diagonal matrices on the left and right), the $(3,k)$-residue refines to

$$\begin{bmatrix} 101 & 101 & 101 & 101 \\ 101 & 10x & 11u & 11p \\ 101 & 11y & 1ev & 1\bar{e}q \\ 101 & 11z & 1\bar{e}w & 1er \end{bmatrix}.$$

The inner products of the first column with the second and third columns give

$$\begin{aligned} 101+10x+11y+11z=000 &\implies x\oplus y\oplus z=1, \\ 101+11u+1ev+1\bar{e}w=000 &\implies u\oplus v\oplus w=1. \end{aligned}$$

The inner product of the second and third columns gives

$$\begin{aligned} &101+11(u\oplus\bar{x})+(1ev+01e+00\bar{y})+(1\bar{e}w+01\bar{e}+00\bar{z}) \\ &=000 \implies x\oplus y\oplus z\oplus u\oplus v\oplus w=1, \text{ a contradiction.} \end{aligned}$$

If $b=c=0$, the inner product of the first and third rows gives $e=h$; similarly, $f=j$. So $\text{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 1d & 1d \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix}.$$

The inner product of the first and third columns gives $d\oplus e\oplus f=0$, so at least one of d,e,f is 0. Up to a row permutation, we may assume $d=0$. So, $\text{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix}.$$

The inner product of the first and third columns implies $e=f$, so left multiplication by K_1 reduces the lde, and moreover, up to permutations of rows and columns, reaching $(1,k-1)$-residue

$$\begin{bmatrix} 1 & x & y & z \\ 0 & \bar{x} & \bar{y} & \bar{z} \\ 1 & u & v & w \\ 0 & \bar{u} & \bar{v} & \bar{w} \end{bmatrix}.$$

Taking the inner product of the first two columns implies $x=u$; similarly, $y=v$ and $z=w$. Depending on whether x,y,z are all 1 or not, the target pattern is (iv) or (iii).

The case $d=g=0$ is similar.

6. *Case (iv)^T*

This case is similar to Case (iv), but a right multiplication by K_1 is needed to decrease the lde.

Thus we have exhausted all cases. The transitions are summarized in Figure 1.

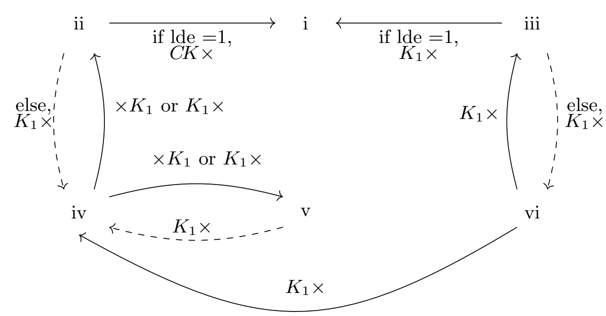

FIG. 1. Transitions between cases. The annotation $K_1\times$ $(\times K_1)$ on an arrow means that if A has the pattern at the tail, then K_1LAR $(LARK_1)$ has the pattern at the head for suitably chosen generalized permutations L and R. Similarly for other annotations. Each of the dashed arrows preserves the lde, each of the rest decreases the lde by 1. Note that CK decomposes into a circuit with two K gates as in Equation (1).

We say a path starting from any case and ending with (i) is *complete*. A complete path decreases the lde of the starting point to 0. For $A\in\mathcal{CCS}$, we define $\text{prkc}(A)$ to be the number of K gates along the complete path starting from A.

Lemma IV.7 (prkc of a complete path)**.** *Given $A\in\mathcal{CCS}$ with lde $l>0$, then* $\text{prkc}(A)$ *satisfies Table II, namely,* $\text{prkc}(A)$ *is determined by l and $\rho_1^l(A)$.*

case	prkc	typical cases
(ii)	$2l$	$ii\to i$, $ii\to iv\to ii\to i$, $ii\to iv\to v\to iv\to ii\to i$
(iii)	$2l-1$	$iii\to i$, $iii\to vi\to iii\to i$
(iv)	$2l-1$	$iv\to ii\to i$, $iv\to ii\to iv\to ii\to i$, $iv\to v\to iv\to ii\to i$
(v)	$2l$	$v\to iv\to ii\to i$, $v\to iv\to v\to iv\to ii\to i$
(vi)	$2l-2$	$vi\to iii\to i$, $vi\to iv\to ii\to i$, $vi\to iii\to vi\to iii\to i$

TABLE II. prkc of a complete path.

Proof. By following the paths. $\square$

Corollary IV.8 (prkc together with residue determines lde)**.** *Given* $A \in \mathcal{CCS}$*, with* $prkc(A) = p$*, if* p *is odd, then* $\mathrm{lde}(A) = (p+1)/2$*; if* p *is even, then* $\mathrm{lde}(A) = p/2$ *provided* $\rho_1^l(A) \neq vi$ *, or* $p/2+1$ *provided* $\rho_1^l(A) = vi$*.*

Lemma IV.9. *For* $A \in \mathcal{CCS}$*,* $\mathrm{prkc}(A) \geq kc(A)$*.*

Proof. We prove this by constructing a circuit C of raw K-count $\mathrm{prkc}(A)$ that implements A. If $\mathrm{lde}(A) = 0$, then A is a generalized permutation. We can find a circuit using zero K gates that implements A as described in Section III C. If $\mathrm{lde}(A) > 0$, by Table II, and merging adjacent generalized permutations as needed, there exist generalized permutations $L_1, L_2, \ldots, L_p, R_1, R_2, \ldots, R_q$ such that

$$\mathrm{lde}(L_p K_1 \cdots L_2 K_1 L_1 A R_1 K_1 R_2 K_1 \cdots R_q) = 0, \text{ i.e.,}$$

there exists a generalized permutation P such that

$$L_p K_1 \cdots L_2 K_1 L_1 A R_1 K_1 R_2 K_1 \cdots R_q = P.$$

Therefore,

$$A = L_1^{-1} i K_1 L_2^{-1} i K_1 \cdots L_p^{-1} P R_q^{-1} i K_1 \cdots R_2^{-1} i K_1 R_1^{-1}.$$

Let C be the right-hand side with each generalized permutation $L_j^{-1} i$ and $R_j^{-1} i$ and $L_p^{-1} P R_q^{-1}$ replaced by a circuit that implements it as described in Section III C. Note $\mathrm{rkc}(C) = \mathrm{prkc}(A)$. □

Note that the proof of Lemma IV.9 gives an algorithm for exact synthesis of a two-qubit Clifford+CS operator A with raw K-count $\mathrm{prkc}(A) = 2\,\mathrm{lde}(A) - \{0, 1, 2\}$. We denote this function by $\mathrm{synth} : \mathcal{CCS} \to \mathcal{CIR}$.

C. Complexity

Since each iteration reduces the lde by 1, using only finitely many operations, and unitaries with lde 0 are generalized permutations, the time complexity of synth is $O(\mathrm{lde}(A))$ for any input $A \in \mathcal{CCS}$.

V. synth IS K-OPTIMAL

To prove that synth is K-optimal, we first prove that the local case transitions and the complete path transitions in Figure 1 are K-optimal. To do so, we first define gate count metrics and other possible ways of decreasing the lde.

Definition V.1. Let D be a Clifford+CS circuit. Recall that $[D]$ is the unitary operator it implements. Define

- $\mathrm{rkc}(D) =$ the number of K gates that D contains.
- $\mathrm{kc}(D) = \min\{\mathrm{rkc}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- $\mathrm{rcs}(D) =$ the number of CS gates that D contains.
- $\mathrm{cs}(D) = \min\{\mathrm{rcs}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- $\mathrm{rlen}(D) =$ the number of gates that D contains.
- $\mathrm{len}(D) = \min\{\mathrm{rlen}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- by abuse of notation, $\mathrm{cs}, \mathrm{kc}, \mathrm{len}$ are also defined on operators that are implementable by circuits.

We say D is K-optimal, CS-optimal, and shortest if $\mathrm{rkc}(D) = \mathrm{kc}(D)$, $\mathrm{rcs}(D) = \mathrm{cs}(D)$, and $\mathrm{rlen}(D) = \mathrm{len}(D)$ respectively. We call $\mathrm{rkc}(D)$ and $\mathrm{rcs}(D)$ the (raw) K-count and (raw) CS-count of D respectively, and $\mathrm{kc}(D)$, $\mathrm{cs}(D)$ the optimal K-count and the optimal CS-count of D respectively.

Definition V.2. An lde-descent (abbreviated as descent) of $A \in \mathcal{CCS}$ is $L, R \in \mathcal{CCS}$ such that $\mathrm{lde}(LAR) \leq \mathrm{lde}(A)$. Since L, R can be written as an alternating sequence of K_1 and generalized permutations (cf. Equation (3)), equivalently, a descent of A is generalized permutations $L_1, L_2, \ldots, L_p, R_1, R_2, \ldots, R_q$ such that $\mathrm{lde}(L_p \cdots K_1 L_2 K_1 L_1 A R_1 K_1 R_2 K_1 \cdots R_q) \leq \mathrm{lde}(A)$, a sub-descent of which is $L_1, L_2, \ldots, L_{p'}, R_1, R_2, \ldots, R_{q'}$ for some $1 \leq p' \leq p$ and $1 \leq q' \leq q$. We call the remaining part of the descent (including all K gates) the exterior. Clearly any path in Figure 1 describes one of the lde-descents; we call it a path descent. A complete path descent is a complete path, i.e., the lde decreases to 0. The K-count of a descent is the number of K gates used in the descent. A K-optimal descent is one of minimum K-count. We abuse the notation of rkc and kc on descents. We write n-descent to emphasize that the lde decrease is n.

Lemma V.3 (Path 1-descent is optimal)**.** *Any path 1-descent for* $A \in \mathcal{CCS}$ *with* $\mathrm{lde}(A) > 0$ *(i.e., an lde-decreasing edge or an lde-preserving edge followed by an lde-decreasing edge in Figure 1) is K-optimal.*

Proof. Let $l = \mathrm{lde}(A)$. This is trivial if $\rho_1^l(A) = (vi)$ or (iv) or $\rho_1^1(A) = (iii)$, since one needs at least one K gate to change the lde, and starting from these cases, one K gate decreases the lde by 1.

For cases $\rho_1^l(A) = (ii)$ or (v), or $\rho_1^l(A) = (iii)$ with $l > 1$, each path has two K gates, which we claim is optimal. This is because the number of ways to use 1 K gate is finite (in $K_1 L_1 A R_1$ or $L_1 A R_1 K_1$, both generalized permutations L_1 and R_1 are finite). We can enumerate them all and check that none of them works. □

Lemma V.4 (K-count 1 1-descent is pattern-locking)**.** *Any K-count 1 1-descent has the same source-target pattern as an undashed edge in Figure 1 prescribes.*

Proof. There are finitely many K-count 1 1-descents from each residue pattern. We can check that the claim holds for each of them. Here we give another proof.

For a K-count 1 1-descent from pattern S to pattern T, we show that the edge $S \to T$ appears in Figure 1. It is easier to consider the reverse, i.e., S is a 1-ascent from

T. If $T = (i)$, then applying an lde-increasing K gate will double the odd entries (over the 1 higher lde); hence $S = (iii)$. If $T = (ii)$, then applying an lde-increasing K gate will double the odd entries; hence $S = (iv)$. If $T = (iii)$, then applying an lde-increasing K gate will double the odd entries; hence $S = (vi)$. If $T = (iv)$, then applying an lde-increasing K gate will double the odd entries; hence $S = (vi)$. If $T = (v)$, then applying an lde-increasing K gate will produce 8 odd entries in two rows (or columns); hence $S = (iv)$. If $T = (vi)$, there is no K-count 1 1-ascent from pattern vi. □

Repeated use of K-optimal 1-descents does not necessarily give an "overall" K-optimal descent for, say, decreasing the lde from 10 to 0. But using the pattern-locking property and the fact that the lde decreases one step at a time (Remark II.10), we can prove that a complete path descent is K-optimal. In other words, there is no shortcut for decreasing the lde, say, from 10 to 0.

Lemma V.5 (The exterior of a K-optimal descent is K-optimal)**.** *For $A \in \mathcal{CCS}$ with lde l, if $L_pK_1 \cdots L_2K_1L_1AR_1K_1R_2K_1 \cdots R_q$ is a K-optimal descent, then any exterior p', q' of it is a K-optimal descent of $L_{p'}K_1 \cdots L_2K_1L_1AR_1K_1R_2K_1 \cdots R_{q'}$.*

Proof. Suppose not; then there exists another descent E of $L_{p'}K_1 \cdots L_2K_1L_1AR_1K_1R_2K_1 \cdots R_{q'}$ with fewer K gates. Replacing the exterior by E gives a descent with fewer K gates, contradicting K-optimality. □

Lemma V.6. *A K-count 1 0-descent from pattern vi cannot reach pattern ii or v.*

Proof. Equivalently, any K-count 1 0-descent from pattern ii or v cannot reach pattern vi. As in the proof of Lemma IV.4, we can, without loss of generality, assume the $(2, l)$-residue of ii and v are

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 01 & 00 \\ 01 & 01 & 0d & 0d \\ 00 & 00 & 0d & 0d \end{bmatrix} \text{ and } \begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 10 & 11 & 10 & 11 \\ 10 & 11 & 11 & 10 \end{bmatrix}.$$

Then any K-count 1 0-descent from them cannot produce pattern vi. Note that this lemma can also be shown by enumeration, since there are finitely many K-count 1 0-descents from pattern vi. □

Lemma V.7 (Complete path descent is optimal)**.** *For $A \in \mathcal{CCS}$ with lde l, the complete path descent of A, i.e., the path l-descent of A, is K-optimal.*

Proof. By induction on $\mathrm{lde}(A)$. If $\mathrm{lde}(A) = 0$, the empty path 0-descent is K-optimal. Suppose for $\mathrm{lde}(A) = l > 0$, the path l-descent from A is K-optimal. For A with $\mathrm{lde}(A) = l + 1$, let $L_pK_1 \cdots L_2K_1L_1AR_1K_1R_2K_1 \cdots R_q$ be a K-optimal $(l + 1)$-descent, for some p, q, and call it D. Then there exists a sub-descent $1 \le p' \le p, 1 \le q' \le q$ of minimal K-count, call it D_1, such that, if we let $M = L_{p'}K_1 \cdots L_2K_1L_1AR_1K_1R_2K_1 \cdots R_{q'}$, $\mathrm{lde}(M) = l$. By the induction hypothesis, the path l-descent starting from M, call it H, is K-optimal, and so is the exterior, call it E, of the sub-descent. Thus, $\mathrm{kc}(E) = \mathrm{rkc}(E) = \mathrm{prkc}(M)$. Since D_1 is of minimal K-count, the K-count 1 exterior E_1 of it decreases the lde from $l + 1$ to l (otherwise, this exterior can be stripped off, contradicting minimality). By pattern-locking, the possible pattern-pairs for E are $iii \to i, iv \to ii, iv \to v, vi \to iii, vi \to iv$. So $\rho_1^l(M) \ne vi$; then $\mathrm{prkc}(M) = 2l, 2l - 1$ by Lemma IV.7. Note that $\mathrm{rkc}(D_1) + \mathrm{rkc}(E) = \mathrm{rkc}(D) < \mathrm{prkc}(A)$, otherwise $\mathrm{prkc}(A)$ is optimal. So $\mathrm{rkc}(D_1) < \max\{2(l + 1) - \{2l, 2l - 1\}\} = 3$. Note $\mathrm{rkc}(D_1) > 0$, since at least one K gate is needed to change the lde.

If $\mathrm{rkc}(D_1) = 1$, by pattern-locking, $\rho_1^{l+1}(A) \to \rho_1^l(M) \in \{iii \to i, iv \to ii, iv \to v, vi \to iii, vi \to iv\}$. If $\rho_1^{l+1}(A) \to \rho_1^l(M) \in \{iii \to i, iv \to ii, iv \to v\}$, then $\mathrm{prkc}(A) = 2(l + 1) - 1 = 1 + 2l = \mathrm{rkc}(D_1) + \mathrm{rkc}(E) = \mathrm{kc}(D)$, i.e., the path $(l + 1)$-descent of A is K-optimal. If $\rho_1^{l+1}(A) \to \rho_1^l(M) \in \{vi \to iii, vi \to iv\}$, then $\mathrm{prkc}(A) = 2(l + 1) - 2 = 1 + (2l - 1) = \mathrm{rkc}(D_1) + \mathrm{rkc}(E)$, i.e., the path $(l + 1)$-descent of A is K-optimal.

If $\mathrm{rkc}(D_1) = 2$, by pattern-locking, the possible pattern-pairs for E_1 are $iii \to i, iv \to ii, iv \to v, vi \to iii, vi \to iv$. If the pattern-pair of E_1 is among $iii \to i, iv \to ii, iv \to v$, then $\mathrm{kc}(D) = \mathrm{rkc}(D_1) + \mathrm{rkc}(E) = 2 + 2l = 2(l + 1) \ge \mathrm{prkc}(A)$, i.e., the path $(l + 1)$-descent of A is K-optimal. If the pattern-pair of E_1 is among $vi \to iii, vi \to iv$, then D_1 is lde-preserving and has pattern-pair $\rho_1^{l+1}(A) \to vi$. The latter forces $\rho_1^{l+1}(A)$ to be not ii or v by Lemma V.6. Thus $\mathrm{rkc}(D_1) + \mathrm{rkc}(E) = 2 + (2l - 1) = 2(l + 1) - 1 \ge \mathrm{prkc}(A)$, i.e., the path $(l + 1)$-descent of A is K-optimal. □

A. Highlights

Corollary V.8. *For any $A \in \mathcal{CCS}$, let $k = \mathrm{prkc}(A)$. Then* synth *outputs a K-optimal circuit C implementing A, using k K gates, at most $k+1$ CS gates, and at most $9(k + 1) + k = 10k + 9$ gates, up to a global phase.*

Proof of Corollary V.8. Suppose synth is not K-optimal; then we can synthesize A with a smaller K-count than $\mathrm{prkc}(A)$, and we can transform this synthesis to a descent of A using fewer K gates, contradicting the K-optimality of the path $\mathrm{lde}(A)$-descent.

The output is an alternating sequence of generalized permutations and K_1 gates as in Equation (3), with the K-count k being optimal. Then there are $k+1$ generalized permutations, each implementable by at most 1 CS gate and at most 9 gates, up to a global phase. Thus the claim holds. □

Theorem V.9. *For any $A \in \mathcal{CCS}$, $\frac{\mathrm{kc}(A)}{2} - 1 \le \mathrm{cs}(A) \le \mathrm{kc}(A) + 1$.*

Proof. Any Clifford+CS operator A can be implemented by an alternating sequence of Clifford parts C_i and a *CS* gate

$$A = C_0 \cdot CS \cdot C_1 \cdot CS \cdot \ldots \cdot C_n. \tag{2}$$

By enumerating all two-qubit Clifford operators, each C_i is implementable by at most 2 K gates. Then we have

$$\mathrm{kc}(A) \leq 2(\mathrm{rcs}(A) + 1) \implies \frac{\mathrm{kc}(A)}{2} - 1 \leq \mathrm{cs}(A).$$

Any Clifford+CS operator A can also be written as an alternating sequence of generalized permutations P_i and K_1 gates (by replacing every K_0 with $Ex \cdot K_1 \cdot Ex$):

$$A = P_0 \cdot K_1 \cdot P_1 \cdot K_1 \cdot \ldots \cdot P_n. \tag{3}$$

Each generalized permutation uses at most 1 CS gate and no K gate, so

$$\mathrm{cs}(A) \leq \mathrm{rkc}(A) + 1 \implies \mathrm{cs}(A) \leq \mathrm{kc}(A) + 1.$$

□

Corollary V.10. *Theorem V.9 implies*

$$\frac{k+1}{\mathrm{cs}(A)} \leq \frac{k+1}{\frac{k}{2} - 1} = 2 + \frac{4}{k-2},$$

that is, the raw CS-count of the output of synth *is at most 2 times the optimal CS-count* cs(A) *asymptotically.*

Later we show that the K-count of Glaudell et al.'s CS-optimal synthesis algorithm is also at most 2 times optimal asymptotically. It seems that for the CS and K gates, achieving optimality of one comes at the cost of losing optimality of the other. But experiments show that our algorithm always achieves the optimal CS-count, while Glaudell et al.'s rarely achieves the optimal K-count.

Remark V.11. Theorem V.9 and Lemma IV.7 imply $l - 2 \leq \mathrm{cs}(A) \leq 2l + 1$, which is similar to Proposition 6.4 of [17], where they have bounds $\frac{l-3}{2} \leq \mathrm{cs}(A) \leq 2l + 2$. Our method improves the lower bound by a factor of 2.

B. K-optimality of Glaudell et al.'s algorithm

Although Glaudell et al. [17] did not consider the K-count, we can derive from their CS-optimality that for any unitary A,

$$\mathrm{cs}(A) - 1 \leq \mathrm{kc}(A) \leq 2\,\mathrm{cs}(A) + 2.$$

This follows because their output has the form of Equation (2) with optimal CS-count, and each Clifford part between two CS gates has K-count 1 or 2. Let E be such a Clifford part. First, $\mathrm{kc}(E) \neq 0$: if E were a generalized permutation (which contains no CS gate since E is Clifford), then the two flanking CS gates in $CS \cdot E \cdot CS$ could be cancelled, contradicting CS-optimality. The cancellation follows by direct calculation over the finitely many generalized permutations. That $\mathrm{kc}(E) \leq 2$ follows from enumerating all two-qubit Cliffords. Thus Glaudell et al.'s algorithm produces circuits with K-count at most 2 times optimal asymptotically.

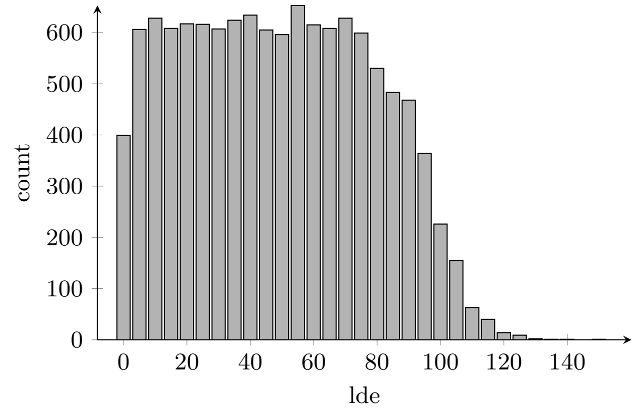

FIG. 2. Distribution of the least denominator exponent (lde) across the 12,000 test operators, binned in groups of 5. The lde ranges from 0 to 152, with a mean of 51 and a median of 50.

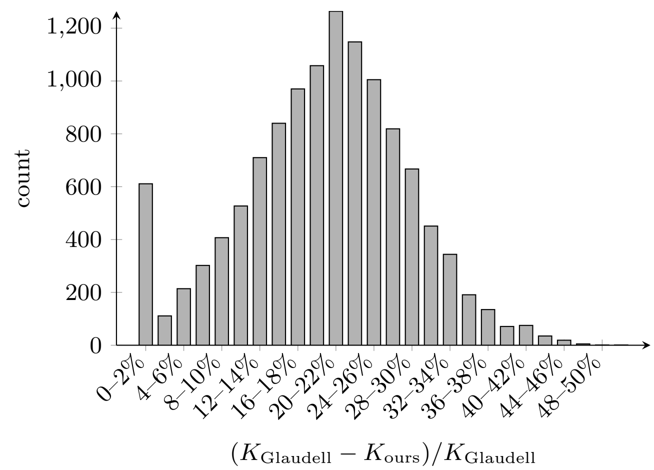

FIG. 3. Distribution of the relative K-count reduction $(K_{\mathrm{Glaudell}} - K_{\mathrm{ours}})/K_{\mathrm{Glaudell}}$, binned in 2 % increments. Our method improves on Glaudell et al.'s K-count in 95.1 % of cases, with a mean reduction of 19.7 %, a median of 20.3 %, and a maximum of 50.0 %.

VI. EXPERIMENT

We implemented our algorithm in Haskell and compared it with the CS-optimal synthesis algorithm of Glaudell et al. [17]. Unitaries with lde values in a given range were randomly generated and fed to both algorithms; the output circuits were compared on rcs and rkc. The output of their algorithm has the form of Equation (2) with optimal CS-count. We augmented their algorithm with one additional step to minimize the K-count: for each C_i, we replace it with an equivalent circuit of minimum K-count (which may be 0, 1, or 2). We randomly generated 12,000 two-qubit Clifford+CS operators with lde ranging from 0 to 152; data are available in Section VI C. Figure 2 shows the distribution of lde values, and Figure 3 shows the distribution of the relative

K-count reduction.

For every operator, our CS-count and that of Glaudell et al. are identical, so our algorithm is also CS-optimal in practice. The K-count is strictly reduced in 95.1 % of cases; see Figure 3.

A. Source of improvement

From the data, the K-count of Glaudell et al. is well approximated by twice the CS-count:

$$K_{\text{Glaudell}} \approx 2 \cdot CS\text{-count}.$$

The data are consistent with Lemma IV.7, i.e., our K-count is well approximated by twice the lde:

$$K_{\text{ours}} \approx 2 \cdot \text{lde}.$$

Then the absolute K-count saving is approximately $2 \cdot (CS\text{-count} - \text{lde})$. The relative reduction is roughly stable across lde values, hovering around 19–21% for lde between 5 and 80, and tapering off at the extremes (possibly due to smaller sample sizes).

B. Worst cases

From Figure 3, for a non-negligible number of operators, namely 588 out of 12,000 (4.9%), our method achieves no K-count reduction. In particular, the following family of circuits:

$$W = CK \cdot KC \cdot CK \cdot KC \cdots$$

is one of the worst cases. One can verify that $\text{lde}(W) = \text{cs}(W)$. For this family of circuits, our algorithm and Glaudell et al.'s produce the same K-count $2 \cdot \text{lde}$.

C. Code and data availability

Code and data are available from [7].

VII. CONCLUSION AND FUTURE WORK

We have given an efficient, K-optimal, CS-near-optimal algorithm for the exact synthesis of two-qubit Clifford+CS circuits from a given unitary matrix. The theoretical worst-case CS-count is 2 times optimal, but in practice our method always achieves the optimal CS-count and consistently achieves a better K-count than the best previously known algorithm.

The algorithm of Li et al. [29] produces circuits whose T-count is at worst 10 times optimal. We believe our method can be adapted to improve this bound. One can also try to optimize the total gate count, as experiments show that the raw length of the output circuit is much below the theoretical bound. One can also try to extend our method to the three-qubit case, though the number of residue patterns is far greater than the 6 in Lemma IV.1, which might cause some difficulty and affect the tightness of the bound. Extending to the four-qubit case is clearly more difficult. Extensions to the two-qubit Clifford+V and Clifford+Cyclotomic gate sets are promising and should be investigated. As experiments show, our method achieved CS-optimality for all 12,000 randomly generated operators. Hence, we conjecture that our method is also CS-optimal, and we leave the proof to future work.

VIII. ACKNOWLEDGMENTS

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